DATA

DATA MINING

BSCS 3A

CHRISTIAN SY

Post Case Study: *“SmartMart’s Customer Insight Initiative”*

Case Background

SmartMart is a medium-sized retail chain operating both physical stores and an online shopping platform. Over the past five years, the company has accumulated a **large database of customer transactions**, loyalty card records, mobile app usage logs, and customer feedback.

Management believes that hidden patterns in this data could help:

- Improve product placement
- Predict customer demand
- Personalize promotions
- Reduce inventory waste

To achieve this, SmartMart plans to adopt **data mining techniques** and has formed a small analytics team composed of IT staff, business analysts, and consultants.

Post Case Study: *“SmartMart’s Customer Insight Initiative”*

However, concerns have been raised regarding:

- The privacy of customer information
- The accuracy and reliability of extracted patterns
- Whether the company fully understands the difference between data mining, statistics, and AI
- The tools and skills needed to responsibly implement the initiative

You are part of an advisory group asked to evaluate the plan.

Post Case Study: *“SmartMart’s Customer Insight Initiative”*

1. Understanding Data Mining

a. Based on the case, explain what data mining means in your own words.

b. Why is data mining more than simply storing or querying data?

2. Data Mining Tasks. Identify which data mining tasks are applicable in SmartMart’s case.

For each, briefly explain how it could be used.

- Classification
- Clustering
- Association Rule Mining
- Prediction / Forecasting
- Anomaly Detection

Post Case Study: *“SmartMart’s Customer Insight Initiative”*

3. Relationship to Other Fields

Explain how SmartMart’s initiative relates to the following fields:

Field	Role in the Case
Statistics	
Optimization	
Machine Learning	
Artificial Intelligence	

4. Tools and Technologies

a. What types of data mining tools would be appropriate for SmartMart (e.g., spreadsheet-based, open-source, enterprise tools)?

b. Why might Python-based tools be preferred over traditional spreadsheet tools as data size grows?

Post Case Study: *“SmartMart’s Customer Insight Initiative”*

5. Data Considerations

a. Identify at least three types of data used in the case.
b. What data quality issues might affect the accuracy of results?
c. Why is data preprocessing critical before applying data mining techniques?

6. Legal, Privacy, and Security Issues

a. What privacy risks arise from analyzing customer behavior data?
b. What ethical boundaries should SmartMart observe when using customer data?
c. How can data mining cross the line into unethical or illegal practice?

7. Decision-Making Reflection

a. Should SmartMart proceed with its data mining initiative? Why or why not?
b. What safeguards should be implemented before deployment?

1. UNDERSTANDING DATA MINING

a. Based on the case, explain what data mining means in your own words.

In SmartMart's case, **data mining** means analyzing large volumes of customer and sales data to discover hidden patterns, relationships, and trends that can support better business decisions.

Rather than just knowing:

- ✓ What was sold
- ✓ When it was sold
- ✓ Who bought it

SmartMart wants to understand:

- ✓ Which customers are likely to respond to promotions
- ✓ What products are often purchased together
- ✓ Which customers are at risk of churning
- ✓ What sales will look like next month

So, data mining transforms raw transaction records into **actionable insights**.

1. UNDERSTANDING DATA MINING

b. Why is data mining more than simply storing or querying data?

Storing and querying data (e.g., using SQL) answers questions like:

- ✓ "How many laptops were sold in January?"
- ✓ "List customers from Region A."

That is **descriptive retrieval**.

Data mining goes further by:

- ✓ Discovering patterns automatically
- ✓ Predicting future outcomes
- ✓ Identifying hidden relationships
- ✓ Detecting unusual behavior

For example:

- ✓ A database query shows total sales.
- ✓ A data mining model predicts next month's demand.

Thus, data mining involves **pattern discovery, modeling, and decision support**, not just retrieval.

2. DATA MINING TASKS

1. Classification. Assigning data into predefined categories.

Classifying customers as:

- ✓ High-value vs. low-value
- ✓ Likely to churn vs. loyal
- ✓ Fraudulent vs. normal transactions

Build a model that predicts whether a customer will respond to a promotional email.

2. Clustering. Grouping similar data points without predefined labels.

Customer segmentation

- ✓ Budget shoppers
- ✓ Brand-loyal customers
- ✓ Seasonal buyers

SmartMart could then design targeted marketing campaigns for each cluster.

2. DATA MINING TASKS

3. Association Rule Mining. Discovering relationships between items.

Classic retail example:

- ✓ Market basket analysis.

Example rule:

"If a customer buys bread and peanut butter, they are likely to buy jam."

SmartMart can:

- ✓ Arrange products strategically
- ✓ Create combo promotions
- ✓ Recommend complementary items

4. Prediction / Forecasting. Estimating future values.

Applications:

- ✓ Sales forecasting
- ✓ Demand prediction
- ✓ Inventory planning
- ✓ Revenue projections

Predicting holiday season demand to avoid stockouts.

2. DATA MINING TASKS

5. Anomaly Detection. Identifying unusual patterns.

Applications:

- ✓ Fraud detection
- ✓ Abnormal purchase behavior
- ✓ Inventory shrinkage detection

A customer suddenly places 20 high-value orders from different locations → possible fraud.

3. RELATIONSHIP TO OTHER FIELDS

1. Statistics. SmartMart uses statistical methods for:

- ✓ Estimating averages
- ✓ Confidence intervals
- ✓ Hypothesis testing
- ✓ Regression modeling

Statistics provides the mathematical foundation for inference and uncertainty measurement.

2. Optimization

Used in:

- ✓ Pricing strategies
- ✓ Inventory control
- ✓ Supply chain logistics
- ✓ Promotion budget allocation

Optimization helps find the **best decision under constraints**.

3. Machine Learning

Machine learning enables:

- ✓ Predictive models
- ✓ Pattern recognition
- ✓ Recommendation systems
- ✓ Automated classification

Data mining heavily relies on ML algorithms like decision trees, clustering, and regression.

4. Artificial Intelligence (AI)

AI extends ML into:

- ✓ Intelligent recommendation engines
- ✓ Chatbots
- ✓ Dynamic pricing systems
- ✓ Automated decision-making systems

Your Role in the Case

As a data professional, your responsibilities would include:

- ✓ Cleaning and preparing data
- ✓ Selecting appropriate models
- ✓ Evaluating model performance
- ✓ Ensuring ethical and legal compliance
- ✓ Translating insights into business recommendations

You serve as the bridge between raw data and strategic decisions.

a. Appropriate Tools for SmartMart

Spreadsheet-based tools (e.g., Excel)

- ✓ Small datasets
- ✓ Basic descriptive analysis

Open-source tools

- ✓ Python (Pandas, Scikit-learn)
- ✓ R
- ✓ Jupyter Notebooks

Enterprise tools

- ✓ SAS
- ✓ IBM SPSS
- ✓ Cloud platforms (AWS, Azure ML)

For large-scale retail data → open-source or enterprise tools are preferable.

b. Why Python-based tools are preferred as data grows

As data grows:

- ✓ Spreadsheets become slow and memory-limited.
- ✓ Manual processing becomes impractical.
- ✓ Automation becomes necessary.

Python-based tools allow:

- ✓ Handling millions of records
- ✓ Automation and pipelines
- ✓ Advanced machine learning models
- ✓ Integration with databases and cloud systems
- ✓ Scalable computation

Spreadsheets are good for exploration; Python is better for scalability and modeling.

5. DATA CONSIDERATIONS

a. Types of Data in SmartMart

- ✓ Transaction data (purchase history)
- ✓ Customer demographic data
- ✓ Time-series data (daily sales)
- ✓ Behavioral data (clickstream)
- ✓ Possibly text data (reviews)

b. Data Quality Issues

- ✓ Missing values
- ✓ Duplicate customer records
- ✓ Inconsistent category labels
- ✓ Outliers in transaction amounts
- ✓ Incorrect timestamps
- ✓ Biased sampling

c. Why Data Preprocessing is Critical

- ✓ Biased predictions
- ✓ Overfitting
- ✓ Misleading patterns
- ✓ Incorrect business decisions

6. LEGAL, PRIVACY, AND SECURITY ISSUES

a. Privacy Risks

Analyzing behavior data may reveal:

- ✓ Personal habits
- ✓ Income level
- ✓ Lifestyle preferences
- ✓ Health-related purchases

b. Ethical Boundaries

- ✓ Obtain informed consent
- ✓ Anonymize personal data
- ✓ Avoid discriminatory profiling
- ✓ Avoid manipulative pricing
- ✓ Be transparent about data use

c. When Data Mining Becomes Unethical or Illegal

- ✓ Selling personal data without consent
- ✓ Targeting vulnerable customers unfairly
- ✓ Discriminatory pricing based on demographics
- ✓ Violating data protection laws

7. DECISION-MAKING REFLECTION

a. Should SmartMart proceed?

Yes — but responsibly.

Benefits:

- ✓ Better decision-making
- ✓ Increased revenue
- ✓ Improved customer satisfaction
- ✓ Efficient inventory management
- ✓ However, it must be ethically implemented.

b. Safeguards Before Deployment

- ✓ Data governance policy
- ✓ Data anonymization
- ✓ Compliance with privacy laws
- ✓ Bias testing of models
- ✓ Security controls
- ✓ Transparent data usage policies
- ✓ Regular auditing of algorithms

DATA

- **Attributes/Features**
- Types of Data Sets
- Data Quality
- Data Preprocessing
- Similarity and Dissimilarity
- Density

WHAT IS DATA?

Collection of data objects and their attributes

- An attribute (in Data Mining and Machine learning often "feature") is a property or characteristic of an object
  - Examples: eye color of a person, temperature, etc.
  - Attribute is also known as variable, field, characteristic
- A collection of attributes describe an object
  - Object is also known as record, point, case, sample, entity, or instance.

Attributes

Objects

TID	Total_Purchase	Payment_Method	Member_Status
001	850.00	GCash	Member
002	320.00	Cash	Non-Member
003	1,250.00	Credit Card	Member
004	150.00	Cash	Non-Member
005	2,100.00	Credit Card	Member

ATTRIBUTE VS ATTRIBUTE VALUES

Attribute

- A **characteristic or feature** describing a data object (column name).

Attribute Value

- The **actual data stored** for that attribute (cell content).

Customer_ID	Age	Height_cm	Height_ft	Member_Status
1001	21	170	5.58	Yes
1002	35	160	5.25	No
1003	28	180	5.91	Yes

SAME ATTRIBUTE, DIFFERENT VALUE REPRESENTATION

Example: Height

Height is the **same attribute**, but it can be measured in:

Centimeters → 170

Feet → 5.58

Both represent the **same real-world characteristic**, just different units.

Attribute	Value Example
Height	170 cm
Height	5.58 ft

DIFFERENT ATTRIBUTES, SAME TYPE OF VALUES

- ✓ Both attributes contain **integers**.
- ✓ But they mean very different things.

Customer_ID	Age
1001	21
1002	35

Attribute	Value Type	Properties
Customer_ID	Integer	Unique identifier, no natural minimum/maximum
Age	Integer	Has realistic bounds (e.g., 0–120 years)

Even though both are integers:

- Age has **semantic meaning and limits**
- ID is just a **label**, not a measurable quantity

JOLLIBEE BUSINESS EXAMPLE

- ✓ Branch_Code may look numeric, but it is categorical
- ✓ Total_Purchase is numeric and meaningful for calculations

Attribute	Example Values	Notes
Order_ID	5001, 5002	Identifier only
Total_Purchase	250, 890	Numeric with meaning
Payment_Method	Cash, GCash	Categorical
Branch_Code	101, 205	Numeric but categorical label

TYPES OF ATTRIBUTES

Attributes (features/variables) can be classified based on **measurement scale** and **data nature**.

There are **two major perspectives**:

1. **Measurement Scale**
   - Nominal
   - Ordinal
   - Interval
   - Ratio

   Nominal, Ordinal: Categorical (Qualitative)

   Interval, Ratio: Quantitative (Numerical)
2. **Data Nature**
   - Categorical (Qualitative)
   - Quantitative (Numerical)

TYPES OF ATTRIBUTES - GENERAL CLASSIFICATION

Measurement Scale	Category Type	Why?
Nominal	**Categorical (Qualitative)**	Values are labels or names. No numeric meaning.
Ordinal	**Categorical (Qualitative)**	Ordered labels, but differences are not measurable.
Interval	**Quantitative (Numerical)**	Numeric values with meaningful differences, but no true zero.
Ratio	**Quantitative (Numerical)**	Numeric values with meaningful differences and true zero.

TYPES OF ATTRIBUTES - NOMINAL ATTRIBUTES

Values are **names or labels**.
There is **no order** and no mathematical meaning.

Examples (General)

- ✓ ID numbers
- ✓ Eye color
- ✓ Zip codes

Jollibee Examples

- ✓ Payment_Method → {Cash, GCash, Credit Card}
- ✓ Branch_Code → {101, 205, 310}
- ✓ Customer_ID → {1001, 1002, 1003}

Even if numbers are used (like Branch_Code), they are just labels.

Data Nature:

- ✓ Categorical
- ✓ Qualitative

TYPES OF ATTRIBUTES - ORDINAL ATTRIBUTES

Values have a **meaningful order**, but differences between values are not numerically measurable.

Examples (General)

- ✓ Rankings (1st, 2nd, 3rd)
- ✓ Grades (A, B, C)
- ✓ Satisfaction level (Low, Medium, High)

Jollibee Examples

- ✓ Customer_Satisfaction → {Poor, Fair, Good, Excellent}
- ✓ Size → {Small, Medium, Large}
- ✓ Spicy_Level → {Mild, Medium, Spicy}

Data Nature:

- ✓ Categorical
- ✓ Qualitative

TYPES OF ATTRIBUTES - INTERVAL ATTRIBUTES

Numeric values where:

- ✓ Order matters
- ✓ Differences are meaningful
- ✓ BUT no true zero point

Examples (General)

- ✓ Temperature in Celsius
- ✓ Calendar dates

Example:

30°C − 20°C = 10°C difference ✓

But 40°C is NOT twice as hot as 20°C ❌

Jollibee Examples

- ✓ Date_of_Visit (2024, 2025)
- ✓ Year_of_Purchase
- ✓ Store temperature in °C

Data Nature:

✓ Quantitative

✓ Numerical

TYPES OF ATTRIBUTES - RATIO ATTRIBUTES

Numeric values where:

- ✓ Order matters
- ✓ Differences are meaningful
- ✓ There is a true zero
- ✓ Ratios make sense

Examples (General)

- ✓ Height
- ✓ Weight
- ✓ Age
- ✓ Income
- ✓ Counts
- ✓ Time duration
- ✓ Temperature in Kelvin

TYPES OF ATTRIBUTES - WHAT "NO TRUE ZERO" REALLY MEANS

A **true zero** means:

Zero = complete absence of the quantity.

A **non-true zero** means:

Zero is just a reference point, not absence.

The Cake Analogy

Scenario 1: Cake Weight (Ratio Scale)

You baked a cake.

- ✓ 0 kg of cake = no cake at all
- ✓ 1 kg of cake = some cake
- ✓ 2 kg of cake = twice as heavy as 1 kg

Here:

- ✓ Zero means complete absence
- ✓ Ratios make sense
- ✓ You can say "twice as much"

➢ This is a **Ratio scale**

Scenario 2: Oven Temperature in Celsius (Interval Scale)

Your oven is set to:

- ✓ 0°C → oven is still warm, not "no heat"
- ✓ 100°C → hot
- ✓ 200°C → hotter

But you cannot say:

- ✓ 200°C is twice as hot as 100°C ❌
- ✓ Because 0°C does NOT mean "no heat."
  It's just a reference point.

➢ This is an **Interval scale**

DISCRETE AND CONTINUOUS ATTRIBUTES

Both are **Quantitative (Numerical)** attributes, but they differ in how values are counted or measured.

Discrete Attributes. A **discrete attribute** can take only **countable, separate values**.

- ✓ Usually whole numbers (integers).
- ✓ You **cannot** have intermediate values between them.

Attribute	Possible Values	Why Discrete?
Number of students in a class	40, 41, 42	You cannot have 40.5 students
Number of orders at Jollibee	120, 121	Orders are counted
Number of computers in a lab	25, 26	Countable
Number of siblings	0, 1, 2	Cannot be 1.7 siblings

Discrete = **Counting**
If you are counting something → it is usually discrete.

DISCRETE AND CONTINUOUS ATTRIBUTES

Both are **Quantitative (Numerical)** attributes, but they differ in how values are counted or measured.

Continuous Attributes. A **continuous attribute** can take **any value within a range**.
It can include decimals and fractions.

Attribute	**Possible Values**	**Why Continuous?**
Height	170 cm, 170.5 cm	Can be measured precisely
Weight	60 kg, 60.25 kg	Has decimals
Time	10.5 seconds	Measured
Temperature	36.5°C	Can vary continuously

Continuous = **Measuring**
If you are measuring something → it is usually continuous.

DATA

- Attributes/Features
- **Types of Data Sets**
- Data Quality
- Data Preprocessing
- Similarity and Dissimilarity
- Density

DATA - TYPES OF DATASETS

Datasets are commonly classified into **three major categories**:

1. **Record Data**
   - Data Matrix
   - Document Data
   - Transaction Data
2. **Graph (Network) Data**
   - World Wide Web
   - Molecular Structures
3. **Ordered Data**
   - Temporal Data
   - Sequential Data
   - Spatial Data
   - Spatio-Temporal Data
   - Genetic Sequence Data
   - Time-Series Data

Each type has different structure and mining techniques.

DATA - RECORD DATA (TABULAR DATA)

- ✓ This is the **most common dataset type**.
- ✓ Data is organized as a **collection of records (rows)**, where each record has multiple **attributes (columns)**.
- ✓ Think of it as a spreadsheet or database table.

RECORD DATA – DATA MATRIX

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute

Structure:

- ✓ Rows → Objects (data instances)
- ✓ Columns → Attributes (features)

Student ID	**Age**	**GPA**	**Program**
001	20	1.75	BSCS
002	21	1.50	BSIT

Used in:

- ✓ Classification
- ✓ Clustering
- ✓ Regression

This is the standard format used in **machine learning**.

RECORD DATA - DOCUMENT DATA

Each record is a **document**, and attributes are usually:

- ✓ each term is a component (attribute) of the vector,
- ✓ the value of each component is the number of times the corresponding term occurs in the document.

Document	**"AI"**	**"Data"**	**"Mining"**
Doc1	3	5	2
Doc2	0	2	4

Used in:

- ✓ Text mining
- ✓ Sentiment analysis
- ✓ Topic modeling

RECORD DATA - TRANSACTION DATA

- ✓ Each record contains a **set of items**.
- ✓ No fixed number of attributes per row.
- ✓ **Example (Market Basket Data):**

Transaction ID	Items Purchased
T1	{Burger, Fries, Coke}
T2	{Chicken, Rice}
T3	{Burger, Coke}

Used in:

- ✓ Association Rule Mining
- ✓ Market Basket Analysis
- ✓ Apriori Algorithm

DATA - GRAPH DATA (NETWORK DATA)

- ✓ Data is represented as **nodes (vertices)** and **edges (relationships)**.
- ✓ Focus is on relationships, not just attributes.

Nodes (Vertices)
Represent objects.
Examples:
- ✓ Person
- ✓ Website
- ✓ Molecule
- ✓ City
- ✓ Microenterprise

Edges (Links)
Represent relationships.
Examples:
- ✓ “follows”
- ✓ “connected to”
- ✓ “bought with”
- ✓ “located near”

Edge Direction
Graphs can be:
•**Undirected** (A — B)
→ Relationship is mutual
Example: “is friend of”
•**Directed** (A → B)
→ Relationship has direction
Example: “follows” on social media

Weighted Graphs
Edges may have weights.
Example:
- ✓ Strength of friendship
- ✓ Number of messages sent
- ✓ Distance between cities
- ✓ Frequency of co-purchase

WORLD WIDE WEB GRAPH

Structure:

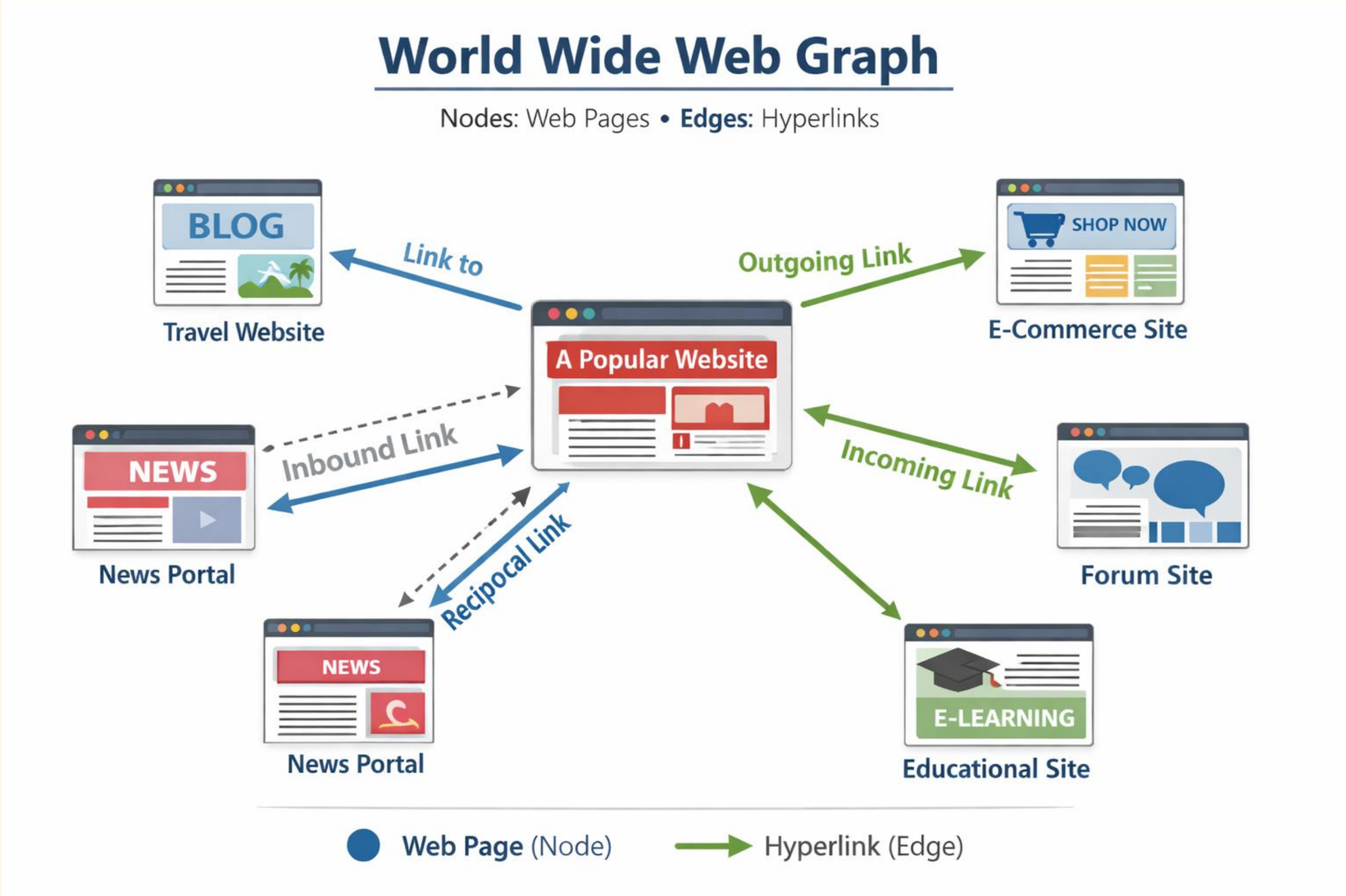
- Nodes → Web pages
- Edges → Hyperlinks

If Page A links to Page B:
A → B

Used in:

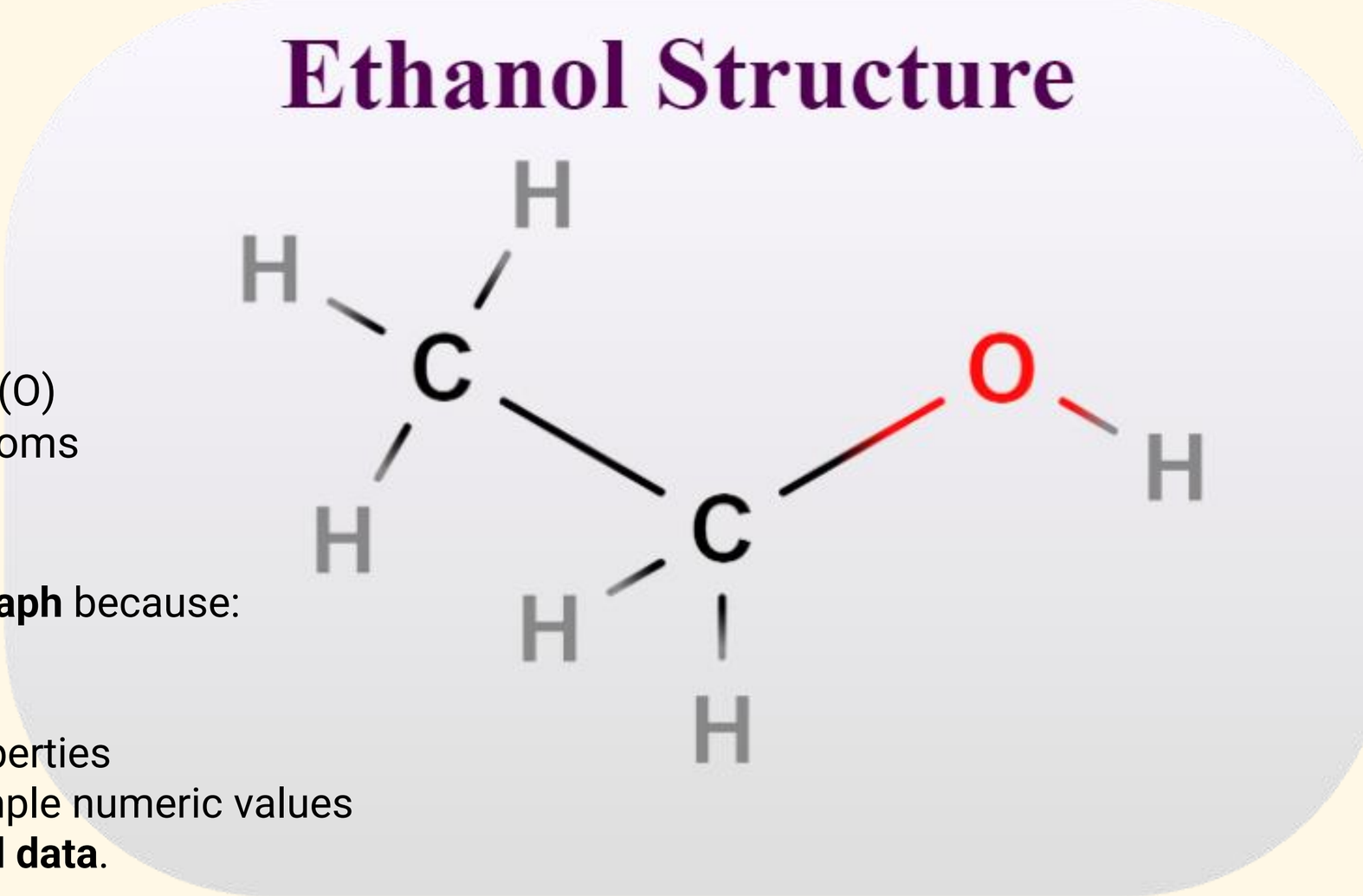
- ✓ Search engine ranking
- ✓ Authority detection
- ✓ Web mining

Famous algorithm:
PageRank



MOLECULAR STRUCTURES

Structure:
Nodes → Atoms
Edges → Chemical bonds
Example:
C — H
C — C
C — O
Here:
Nodes = Carbon (C), Hydrogen (H), Oxygen (O)
Edges = Single or double bonds between atoms



Why Molecular Structures Are Graph Data
A molecule is naturally represented as a **graph** because:

- ✓ Atoms are entities (nodes)
- ✓ Bonds are relationships (edges)
- ✓ Structure determines chemical properties
- ✓ Connectivity matters more than simple numeric values

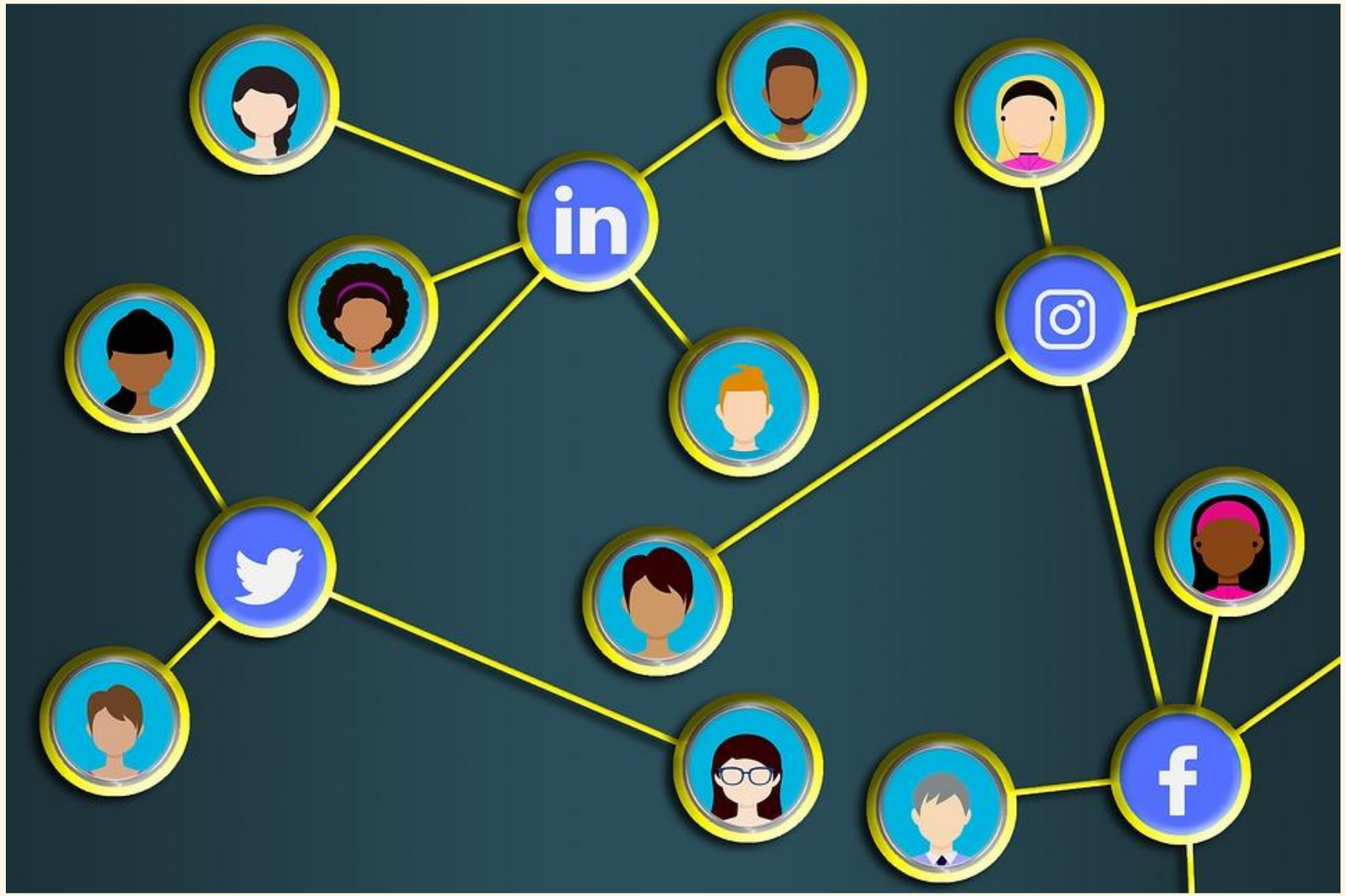
This is not just record data — it is **relational data**.

SOCIAL NETWORK GRAPH

Structure:
Nodes → People
Edges → Friendships
Example:
- A — B
- A — C
- B — D

Used in:
- ✓ Friend recommendation
- ✓ Influencer detection
- ✓ Community detection

Algorithm examples:
- ✓ PageRank
- ✓ Centrality measures
- ✓ Louvain clustering



TRANSPORTATION NETWORK

Structure:

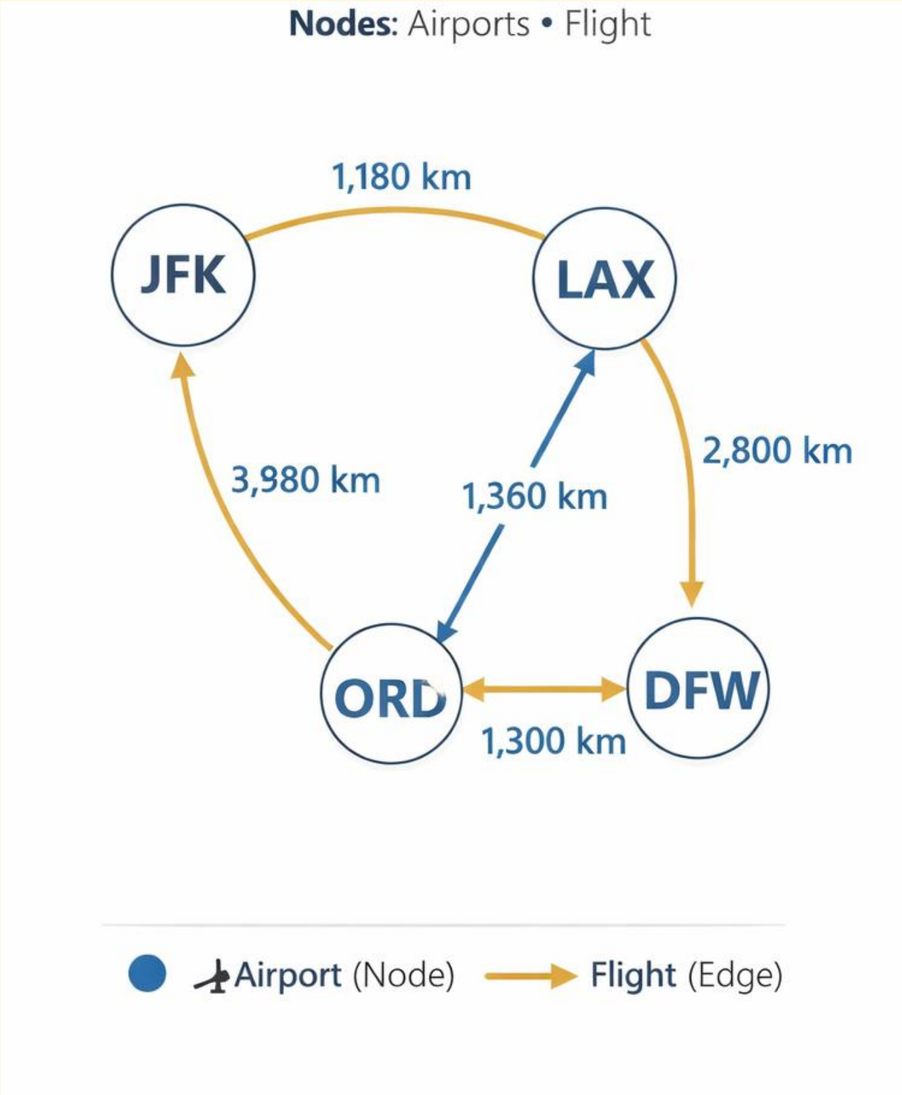
- Nodes → Airports or cities
- Edges → Flights or roads

Edges may have weights:

- ✓ Distance
- ✓ Travel time
- ✓ Cost

Used in:

- ✓ Shortest path algorithms (Dijkstra)
- ✓ Traffic optimization
- ✓ Route planning



E-COMMERCE PRODUCT NETWORK

Structure:

✓ Nodes → Products

✓ Edge → “Frequently bought together”

Example:

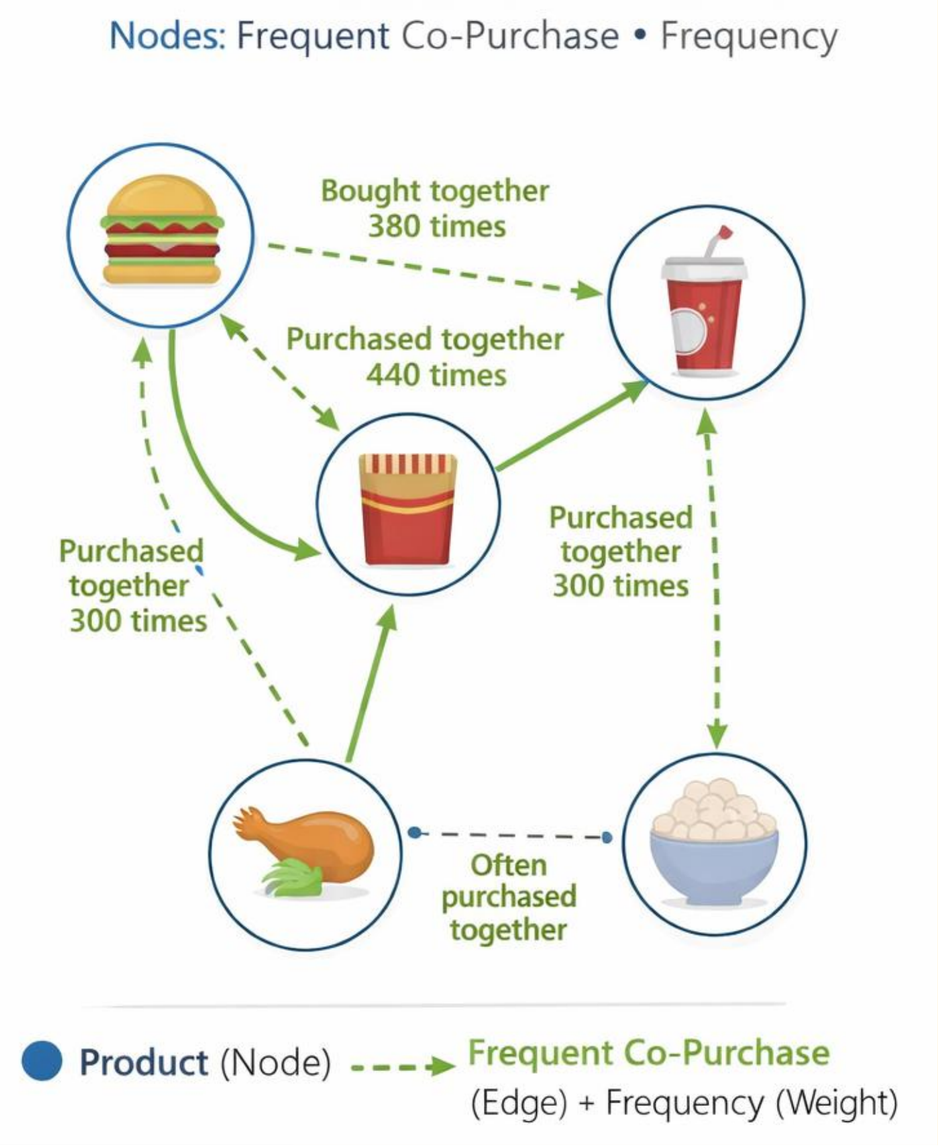
✓ Burger — Fries

✓ Burger — Coke

Used in:

✓ Recommendation systems

✓ Association rule mining (graph version)



ORDERED DATA

Ordered data is a type of dataset where:

- ✓ The order of the objects (records) matters.
- ✓ Unlike simple tabular data (data matrix), where rows are independent, **ordered data has natural sequencing or structure**.
- ✓ If you change the order, the meaning changes.

MAIN TYPES OF ORDERED DATA DATA

Ordered data can be classified into:

- Temporal Data
- Sequential Data
- Spatial Data
- Spatio-Temporal Data
- Genetic Sequence Data
- Time-Series Data

TEMPORAL DATA (TIME-ORDERED DATA)

Data ordered by time.

Date	Sales
Jan	10,000
Feb	12,000
Mar	9,500

If you shuffle the months, trend analysis becomes meaningless.

Used in:

- ✓ Forecasting
- ✓ Trend analysis
- ✓ Anomaly detection

SEQUENTIAL DATA (EVENT-ORDERED DATA)

Data ordered by event sequence (not necessarily exact time).

Customer actions:
Login → View Product → Add to Cart → Checkout

If you change the order:
Checkout → Login ❌ (not logical)

Used in:

- ✓ Clickstream analysis
- ✓ Behavior modeling
- ✓ Recommendation systems

SPATIAL DATA (LOCATION-ORDERED DATA)

Data organized by physical location.

City	Rainfall
Manila	200 mm
Cebu	150 mm

Locations have spatial relationships:

- ➢ Near
- ➢ Adjacent
- ➢ Within region

Used in:

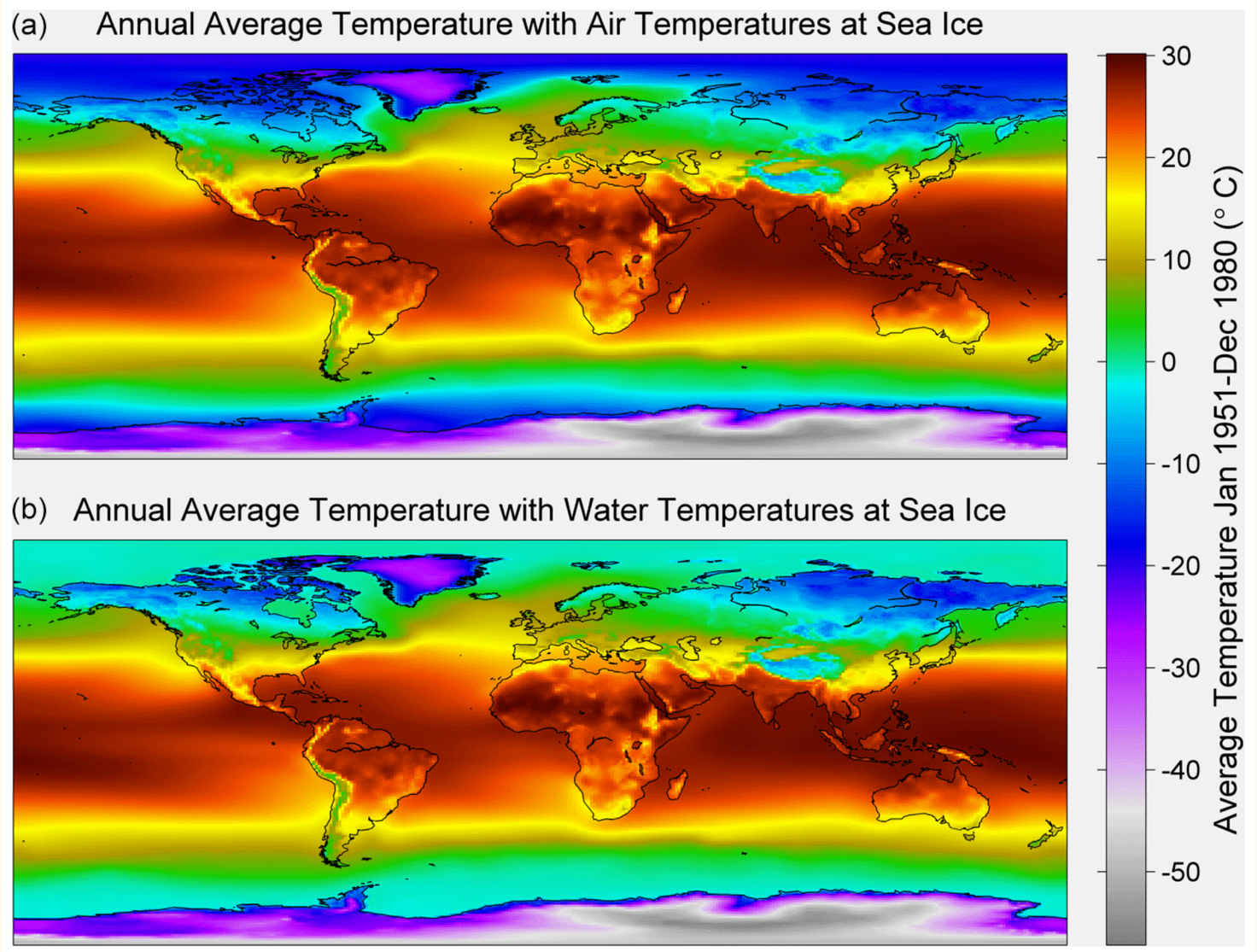
- ✓ GIS systems
- ✓ Environmental monitoring
- ✓ Urban planning

ORDERED DATA: SPATIO-TEMPORAL

Spatio-temporal data is **data that has both:**

- ✓ **Spatial component** → location (where)
- ✓ **Temporal component** → time (when)

It describes **how something changes over space and time**.



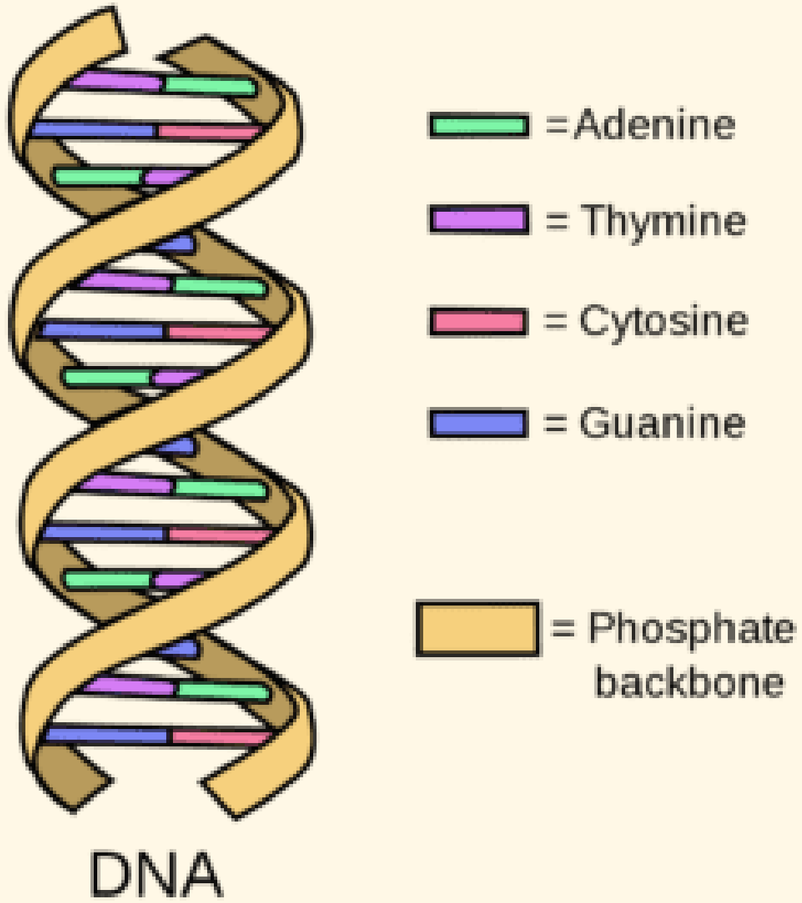
GENETIC SEQUENCE DATA

Biological sequences where order is critical.

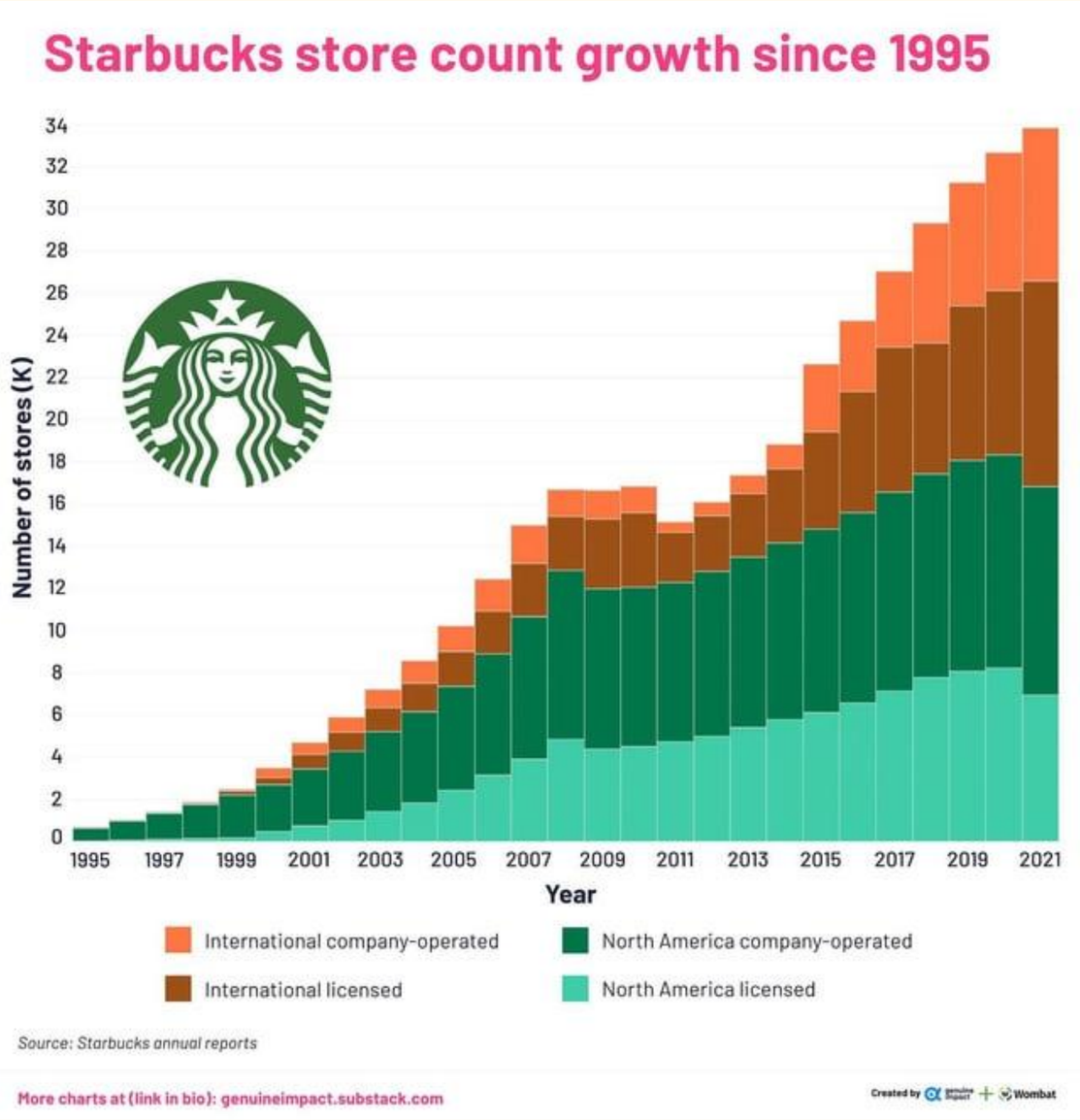
Example:
DNA sequence:
A T C G G T A

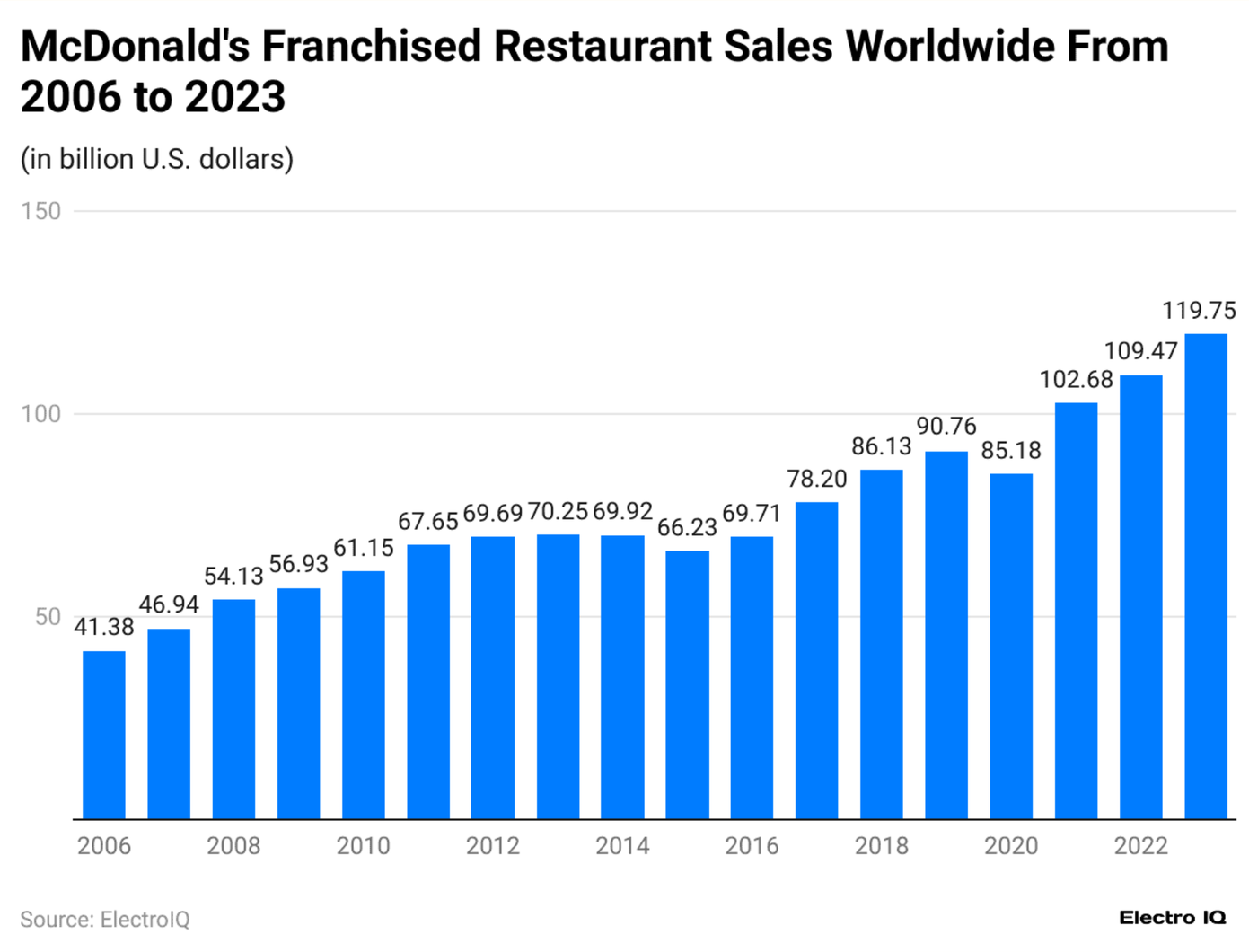
If rearranged:
T A G C G A T → Different meaning
Used in:

- ✓ Bioinformatics
- ✓ Gene pattern discovery



ORDERED DATA: TIME SERIES DATA





WHY GRAPH IS POWERFUL

GRAPH DATA CAPTURES RELATIONSHIPS

THIS ALLOWS:

- ✓ Relationship analysis
- ✓ Influence detection
- ✓ Community discovery
- ✓ Fraud detection
- ✓ Pattern discovery

DATA

- Attributes/Features
- Types of Data Sets
- **Data Quality**
- Data Preprocessing
- Similarity and Dissimilarity
- Density

DATA QUALITY

Data quality refers to - How accurate, complete, consistent, reliable, and usable data is for analysis.

Poor data quality leads to:

- ✓ Wrong predictions
- ✓ Misleading patterns
- ✓ Biased models
- ✓ Poor decision-making

Garbage in → Garbage out.

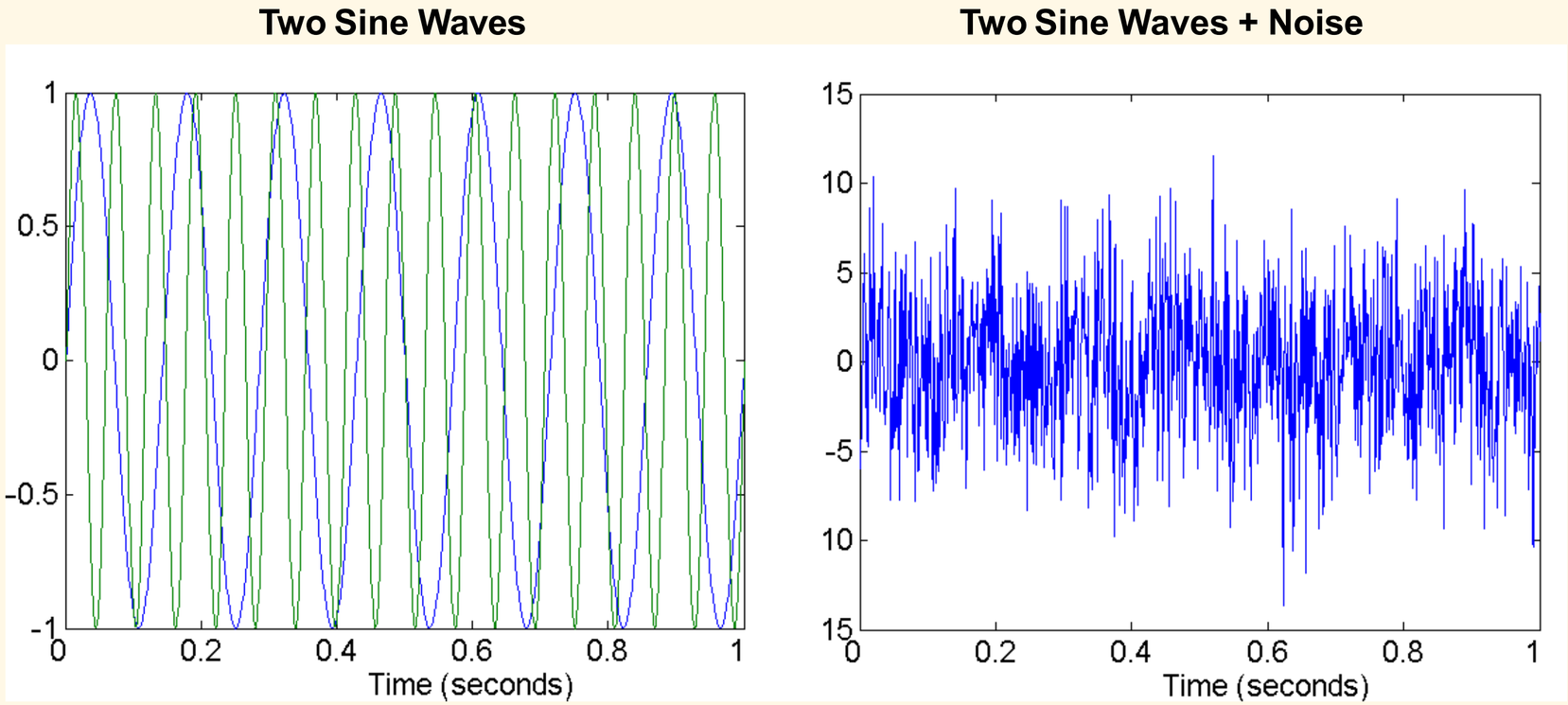
MAJOR TYPES OF DATA QUALITY PROBLEMS

- ✓ Noise and Outliers
- ✓ Missing Values
- ✓ Duplicate Data
- ✓ Inconsistent Data
- ✓ Invalid Data
- ✓ Incomplete Data
- ✓ Data Integration Issues

NOISE

- Noise refers to modification of original values

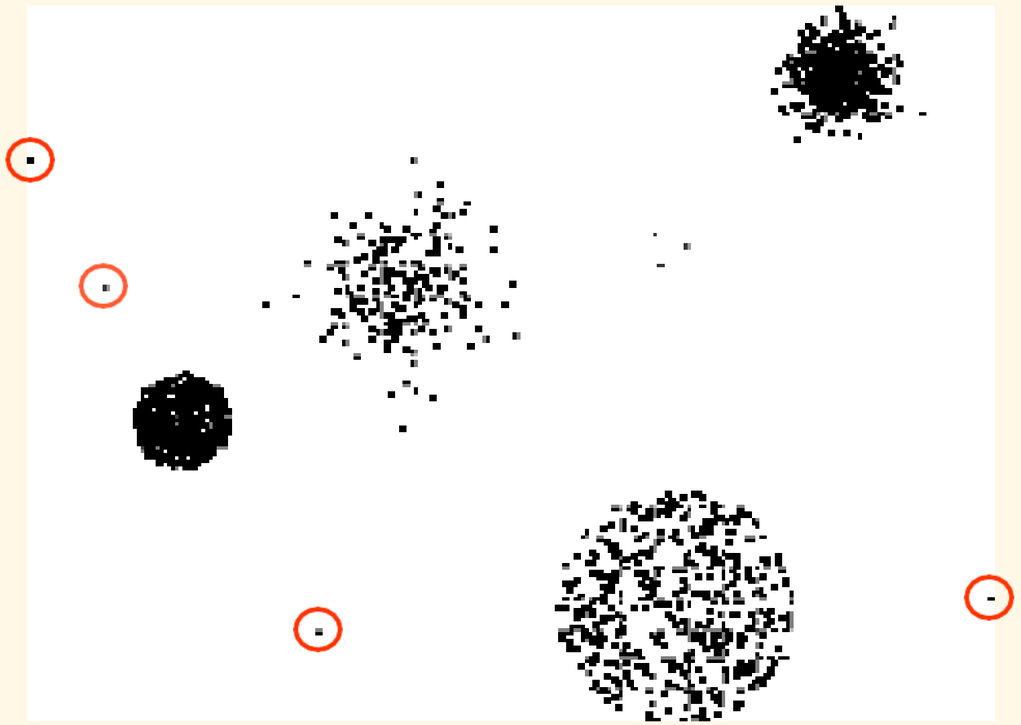
  Examples: distortion of a person's voice when talking on a poor phone, "snow" on television screen, measurement errors.



- Find less noisy data
- De-noise (signal processing)

OUTLIERS

- Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set



Example (Monthly Income):

- 25,000
- 30,000
- 28,000
- 29,000
- 2,500,000 ← Outlier

Outliers can be:

- ✓ Legitimate (e.g., millionaire)
- ✓ Error (extra zero typed)

- Outlier detection + remove outliers

MISSING VALUES

- Reasons for missing values
  - ✓ Information is not collected (e.g., people decline to give their age and weight)
  - ✓ Attributes may not be applicable to all cases (e.g., annual income is not applicable to children)

- Handling missing values
  - ✓ Eliminate data objects with missing value
  - ✓ Eliminate feature with missing values
  - ✓ Ignore the missing value during analysis
  - ✓ Estimate missing values = Imputation

    (e.g., replace with mean or weighted mean where all possible values are weighted by their probabilities)

Age	Income
21	25,000
NULL	30,000
23	NULL

MISSING VALUES - IMPUTATION

Method	Example
Mean	Replace missing Age with average
Median	Good for skewed data
Mode	For categorical data
Regression	Predict missing values
KNN	Fill using nearest neighbors

MISSING VALUES - IMPUTATION

Transaction ID	Branch	Customer Age	Order Type	Total Amount (₱)	Items Purchased	Waiting Time (min)
T001	Manila	22	Dine-in	250	3	8
T002	Cebu	35	Takeout	180	2	5
T003	Manila	?	Dine-in	320	4	10
T004	Davao	28	Delivery	?	5	30
T005	Cebu	40	Takeout	200	?	6
T006	Manila	30	Dine-in	260	3	?

We observe several missing values:

- ✓ Missing age (T003)
- ✓ Missing total amount (T004)
- ✓ Missing items purchased (T005)
- ✓ Missing waiting time (T006)

MISSING VALUES - MEAN IMPUTATION

Transaction ID	Branch	Customer Age	Order Type	Total Amount (₱)	Items Purchased	Waiting Time (min)
T001	Manila	22	Dine-in	250	3	8
T002	Cebu	35	Takeout	180	2	5
T003	Manila	?	Dine-in	320	4	10
T004	Davao	28	Delivery	?	5	30
T005	Cebu	40	Takeout	200	?	6
T006	Manila	30	Dine-in	260	3	?

Missing: Customer Age (T003)
Known ages: 22, 35, 28, 40, 30
Mean age: (22 + 35 + 28 + 40 + 30) / 5
= 155 / 5
= 31
Replace: T003 Age = 31

MISSING VALUES - MEDIAN IMPUTATION

Transaction ID	Branch	Customer Age	Order Type	Total Amount (₱)	Items Purchased	Waiting Time (min)
T001	Manila	22	Dine-in	250	3	8
T002	Cebu	35	Takeout	180	2	5
T003	Manila	?	Dine-in	320	4	10
T004	Davao	28	Delivery	?	5	30
T005	Cebu	40	Takeout	200	?	6
T006	Manila	30	Dine-in	260	3	?

Sorted ages: 22, 28, 30, 35, 40
Median = 30
Replace: T003 Age = 30
✓ Better if age distribution is skewed.

MISSING VALUES - MODE IMPUTATION (CATEGORICAL EXAMPLE)

Transaction ID	Branch	Customer Age	Order Type	Total Amount (₱)	Items Purchased	Waiting Time (min)
T001	Manila	22	Dine-in	250	3	8
T002	Cebu	35	Takeout	180	2	5
T003	Manila	?	Dine-in	320	4	10
T004	Davao	28	Delivery	?	5	30
T005	Cebu	40	Takeout	200	?	6
T006	Manila	30	Dine-in	260	3	?

If Order Type were missing:
Order types: Dine-in, Takeout, Dine-in, Delivery, Takeout, Dine-in
Mode = Dine-in
Replace missing order type with "Dine-in".
✓ Used for nominal data.

MISSING VALUES - REGRESSION IMPUTATION

Transaction ID	Branch	Customer Age	Order Type	Total Amount (₱)	Items Purchased	Waiting Time (min)
T001	Manila	22	Dine-in	250	3	8
T002	Cebu	35	Takeout	180	2	5
T003	Manila	?	Dine-in	320	4	10
T004	Davao	28	Delivery	?	5	30
T005	Cebu	40	Takeout	200	?	6
T006	Manila	30	Dine-in	260	3	?

Missing: Total Amount (T004)
We notice: More items purchased → Higher total amount
Build simple regression model: Total Amount = 50 × (Items Purchased)
For T004: Items = 5
Predicted: 50 × 5 = 250
So: T004 Total Amount ≈ 250
✓ Uses relationships between variables
✓ More realistic than simple mean

MISSING VALUES - KNN IMPUTATION

Transaction ID	Branch	Customer Age	Order Type	Total Amount (₱)	Items Purchased	Waiting Time (min)
T001	Manila	22	Dine-in	250	3	8
T002	Cebu	35	Takeout	180	2	5
T003	Manila	?	Dine-in	320	4	10
T004	Davao	28	Delivery	?	5	30
T005	Cebu	40	Takeout	200	?	6
T006	Manila	30	Dine-in	260	3	?

Missing: Waiting Time (T006)
Compare T006:
Branch: Manila
Order Type: Dine-in
Items: 3
Amount: 260
Most similar transactions:

T001:
Manila
Dine-in
3 items
Waiting time = 8

T003:
Manila
Dine-in
4 items
Waiting time = 10
Average: (8 + 10) / 2 = 9

Replace: T006 Waiting Time = 9

DUPLICATE DATA

Data set may include data objects that are duplicates, or "close duplicates" of one another

- ✓ Major issue when merging data from heterogeneous sources

Examples:

Same person with multiple email addresses

Data cleaning

- ✓ Process of dealing with duplicate data issues
- ✓ ETL tools typically support deduplication

What Can We Do?

- ✓ Remove exact duplicates
- ✓ Merge partial duplicates
- ✓ Standardize format before matching

Example:
“Juan Dela Cruz” vs “Juan D. Cruz”

OTHER COMMON DATA QUALITY PROBLEMS

Inconsistent Data
Example:
Gender column:
Male, male, M, m
Solution:
Standardize format.

Invalid Data
Example:
Age = -5
Date = 32/13/2024
Solution:
Apply validation rules.

Incomplete Data
Example:
Missing key attributes for many rows.
Solution:
Recollect or discard dataset.

WHY DATA QUALITY IS CRITICAL

Poor data quality affects:

- ✓ Accuracy
- ✓ Model performance
- ✓ Business decisions
- ✓ Forecasting reliability

Often:

70–80% of real-world data science work is data cleaning.

PART I - POST ASSESSMENT

CASE STUDY: Smart Healthcare Monitoring System
A hospital implements a smart monitoring system that collects patient data from wearable devices and hospital records.
The dataset includes:

- Patient ID
- Age
- Gender
- Blood Pressure (mmHg)
- Heart Rate (bpm)
- Body Temperature (°C)
- Diagnosis Category (Cardiac, Respiratory, Neurological, etc.)
- Admission Date
- Discharge Date
- ICU Admission (Yes/No)
- Daily activity logs from wearable sensors
- Medical history notes (text data)

PART I - POST ASSESSMENT

Problems observed:

- ✓ Missing blood pressure readings
- ✓ Inconsistent gender coding ("M", "Male", "F", "Female")
- ✓ Duplicate patient records
- ✓ Extreme temperature values (e.g., 150°C)
- ✓ Some patients have incomplete medical history

PART I - POST ASSESSMENT

PART I – ATTRIBUTES / FEATURES

Q1. Identify the attribute types in the dataset.
Classify each as:

- ✓ Nominal
- ✓ Ordinal
- ✓ Interval
- ✓ Ratio
- ✓ Numerical (Discrete/Continuous)

Q2. Distinguish between:

a. Discrete and continuous attributes
b. Nominal and ordinal attributes
c. Interval and ratio attributes

Provide examples from the case study.

PART I - POST ASSESSMENT

PART I – ATTRIBUTES / FEATURES

Q3. Identify possible:

- ✓ Identifier attribute
- ✓ Predictive attributes
- ✓ Target variable (if predicting ICU admission)

Explain your reasoning.

PART I - POST ASSESSMENT

PART II – TYPES OF DATA SETS

Q4. Identify the types of datasets present in this case:

- ✓ Record data
- ✓ Time-series data
- ✓ Transaction data
- ✓ Text data
- ✓ Spatial data
- ✓ Graph/network data

Explain your answers.

PART I - POST ASSESSMENT

PART II – TYPES OF DATA SETS

Q5. If wearable devices record heart rate every minute, what type of dataset does this become?
How is it different from simple record data?

PART I - POST ASSESSMENT

PART III – DATA QUALITY

Q6. Identify at least five data quality issues in this dataset (provide your own dataset).

Categorize them as:

- ✓ Missing data
- ✓ Noisy data
- ✓ Inconsistent data
- ✓ Duplicate data
- ✓ Outliers

PART I - POST ASSESSMENT

PART III – DATA QUALITY

Q7. Explain the impact of poor data quality on:

- ✓ Model accuracy
- ✓ Bias
- ✓ Interpretability
- ✓ Patient safety

THANK YOU!

ANY QUESTIONS?